ACLS

UPDATE

2026

Advanced Cardiovascular Life Support

American Heart Association
Guidelines — 2026 Edition

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Presenter

Comprehensive

2026

Latest Guidelines

Evidence-Based • Updated Protocols • Clinical Excellence

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What's New in 2026

CPR Compression Depth

Updated target 2.0–2.4" (5–6 cm); avoid >2.4" to prevent rib fractures

Amiodarone vs Lidocaine

Lidocaine now equivalent first-line option; both acceptable for shock-refractory VF/pVT

TTM Target Temp Revised

New target 36°C (was 32–36°C); evidence shows no benefit of lower temps

Ventilation Rate

1 breath/6 sec (10/min) during CPR; avoid hyperventilation strictly

ECMO-CPR (ECPR)

Expanded indications for refractory cardiac arrest in eligible centers

POCUS Integration

Bedside ultrasound formally incorporated in arrest algorithm

2026 Guideline Changes at a Glance

Parameter	2020 Guidelines	2026 Update	Class of Evidence
Compression depth	2–2.4 in (5–6 cm)	2.0–2.4 in; avoid >2.4 in	Class I, LOE B-R
Compression rate	100–120/min	100–120/min (unchanged)	Class I, LOE B-R
Ventilation rate (CPR)	1 breath/5–6 sec	1 breath/6 sec (10/min)	Class IIa, LOE C-EO
TTM target	32–36°C for 24 h	36°C for 24 h	Class IIa, LOE B-R
Primary drug VF/pVT	Amiodarone preferred	Amiodarone or Lidocaine	Class IIb, LOE B-R
ECPR (ECMO-CPR)	Selected centers	Expanded criteria/indications	Class IIa, LOE C-LD
Vasopressor timing	ASAP after rhythm check	First dose within 5 min	Class I, LOE B-NR
Capnography	Recommended	Mandatory monitoring	Class I, LOE B-NR

Understanding Evidence Classes

Class I	STRONG	Benefit >>> Risk. Procedure/treatment SHOULD be performed
Class IIa	MODERATE	Benefit >> Risk. It is REASONABLE to perform
Class IIb	WEAK	Benefit ≥ Risk. Procedure/treatment MAY be considered
Class III	HARM	Risk ≥ Benefit. No benefit or potentially harmful. Do NOT perform

Level of Evidence (LOE): A = Multiple RCTs | B-R = Limited RCTs | B-NR = Non-randomized studies | C-LD = Limited data | C-EO = Expert opinion

Adult Cardiac Arrest Algorithm 2026

Unresponsive / Not Breathing Normally
Activate EMS · Get AED · Begin CPR

CPR 30:2 · Attach Monitor/Defibrillator

Rhythm Check — Shockable?

NO

NO — Non-Shockable
Asystole / PEA
Epinephrine ASAP, CPR 2 min

YES

YES — Shockable
VF / Pulseless VT
Defibrillate immediately

After each 2-min CPR cycle: Reassess rhythm → If still shockable: Shock → Epinephrine 1 mg IV/IO q3–5min → Amiodarone 300 mg or Lidocaine 1.5 mg/kg (1st dose) after 3rd shock

ROSC → Post-Arrest Care

No ROSC → Continue / Consider
ECPR / Termination

Shockable vs Non-Shockable Rhythms

SHOCKABLE

Rhythms:

→ Ventricular Fibrillation (VF)

→ Pulseless Ventricular Tachycardia (pVT)

Management:

- Immediate defibrillation
- CPR 2 min after shock
- Epinephrine after 2nd shock
- Amiodarone / Lidocaine after 3rd shock
- Consider reversible causes (Hs & Ts)

NON-SHOCKABLE

Rhythms:

→ Asystole

→ Pulseless Electrical Activity (PEA)

Management:

- Epinephrine 1 mg IV/IO ASAP
- CPR 2 min cycles
- Identify & treat Hs & Ts
- POCUS to guide management
- No routine atropine

Reversible Causes — The Hs & Ts

The 5 H's

Hypovolemia

Tx: IV fluids bolus

Hypoxia

Tx: 100% O_2 / airway

Hydrogen ion (Acidosis)

Tx: $NaHCO_3$ if severe

Hypo/Hyperkalemia

Tx: Calcium / Kayexalate

Hypothermia

Tx: Active rewarming

The 5 T's

Tension Pneumothorax

Tx: Needle decompression

Tamponade (Cardiac)

Tx: Pericardiocentesis

Toxins

Tx: Antidote / supportive

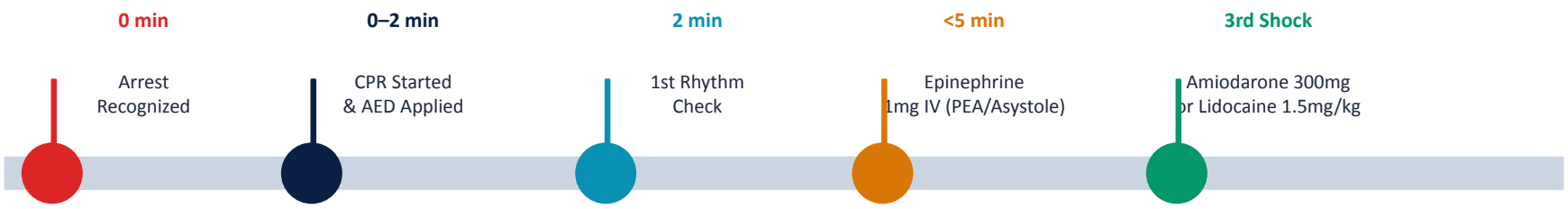
Thrombosis (Pulmonary)

Tx: Thrombolysis / ECPR

Thrombosis (Coronary)

Tx: PCI / thrombolysis

Drug Timing in Cardiac Arrest



Epinephrine: 1 mg IV/IO every 3–5 min throughout arrest

Amiodarone 300 mg IV/IO (2nd dose: 150 mg) OR Lidocaine 1.5 mg/kg (2nd: 0.75 mg/kg)

Sodium Bicarbonate: Only for specific indications (hyperkalemia, TCA OD, prolonged arrest)

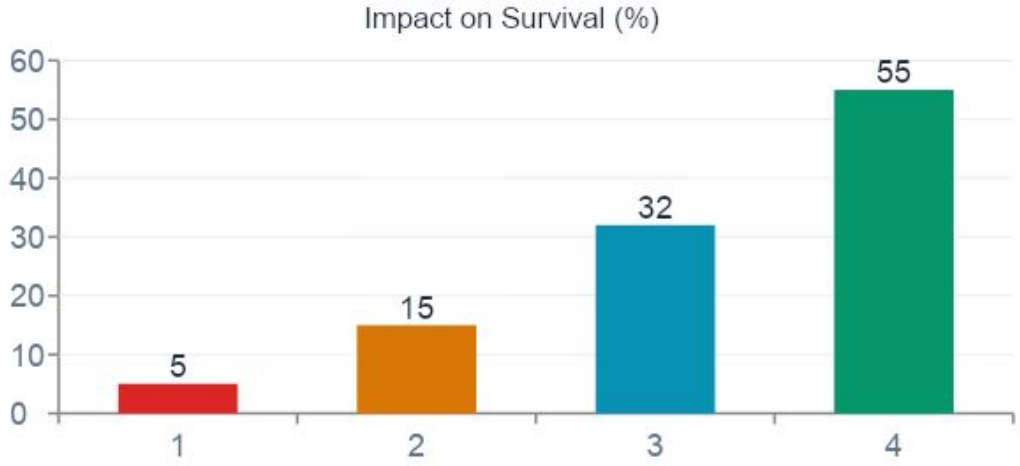
High-Quality CPR — 2026 Standards

100–120
/min
Compression Rate

2.0–2.4
inches
Compression Depth

10
breath/min
Ventilation Rate

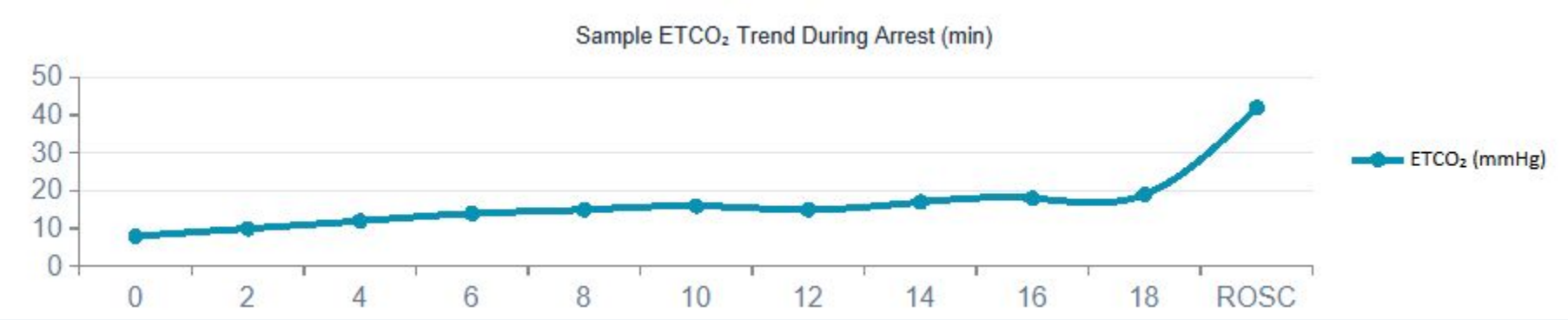
<10
seconds
Max Pause (Pre-shock)



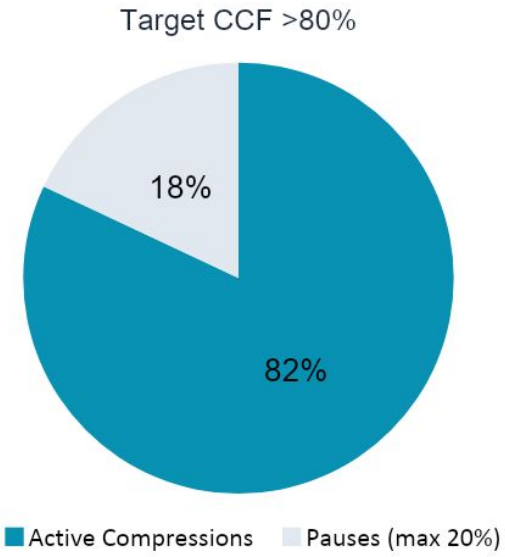
- Full chest recoil between compressions — avoid leaning
- Minimize interruptions — CCF (Chest Compression Fraction) >80%
- Switch compressors every 2 min to prevent fatigue
- Use feedback devices when available (CPR quality monitors)

Capnography in Cardiac Arrest (2026: Class I)

$ETCO_2$ Range	Interpretation	Action
<10 mmHg	Poor CPR quality / Very low perfusion	Improve compression quality
10–20 mmHg	Moderate CPR quality	Optimize compressions, rate & depth
>20 mmHg	Good CPR quality / Consider ROSC	Continue; check pulse if sudden rise
Sudden ↑ to >40	ROSC indicator	Check pulse immediately



Chest Compression Fraction & Feedback Devices



- CCF (Chest Compression Fraction) target: >80%
- Each pause >10 sec markedly reduces cerebral perfusion
- Pre-shock pause should be <10 seconds
- Peri-shock pause (pre + post) should be <20 sec
- Real-time feedback devices: Class IIa recommendation
- Accelerometer-based devices improve depth accuracy
- Audio feedback metronomes help maintain 100–120/min

Defibrillation — 2026 Protocols

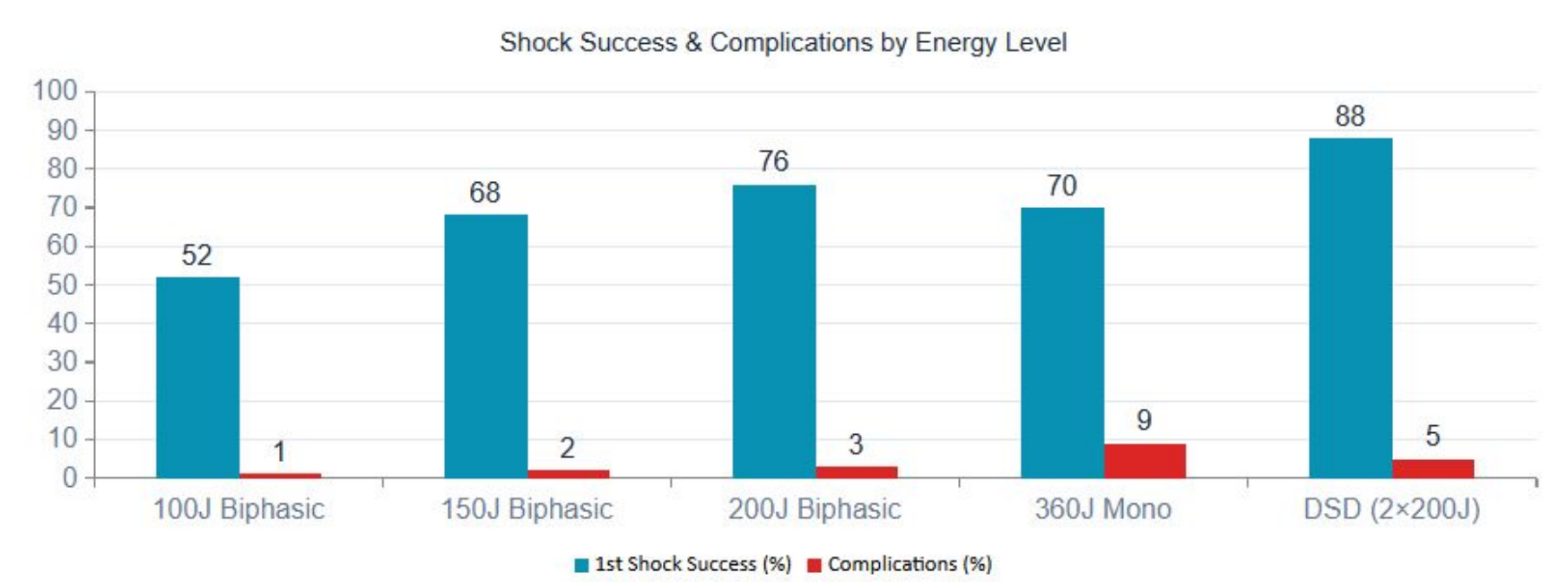
Device Type	Initial Energy	Subsequent Energy	Notes
Biphasic (manufacturer specified)	120–200 J	Same or escalating	Preferred — Class I
Biphasic (if unknown dose)	200 J	200 J or escalating	Use maximum if unknown
Monophasic	360 J	360 J	Fixed energy — less preferred
AED (Automated)	Device-specific	Device-specific	Follow voice prompts
Double Sequential Defib (DSD)	200 J × 2	200 J × 2	Refractory VF — Class IIb (NEW)

NEW 2026: Double Sequential Defibrillation (DSD) — Apply two defibrillators simultaneously for refractory VF after ≥3 standard shocks have failed. Class IIb, LOE C-LD.

Standard Pad Placement:
- Anterior-Lateral (default)
- Anterior-Posterior (preferred for WPW, LVH)

Post-Shock:
- Resume CPR IMMEDIATELY — do not delay
- <10 sec pre-shock pause is mandatory

Defibrillation Energy — Comparative Data



** DSD data from observational studies only. Standard biphasic 200J remains Class I first-line. DSD reserved for refractory VF (Class IIb).*

AED in Public Access Defibrillation

1

Power ON the AED

Open lid or press power button — voice prompts guide rescuer

2

Attach Pads

Anterior-lateral placement; shave excessive hair; dry chest

3

Analyze Rhythm

Clear patient — no touching; AED auto-analyzes rhythm

4

Deliver Shock (if advised)

Clear patient; press SHOCK — immediately resume CPR

5

Resume CPR for 2 min

Begin 30:2 cycles; AED re-analyzes every 2 min

Key Update 2026: AED use by bystanders before EMS arrival improves survival by 3–4×. Deploy within 3–5 min of collapse for best outcomes.

Airway Management in Cardiac Arrest

1

Bag-Mask Ventilation (BVM)

Class I

Indication: Initial airway (all providers)

✓ Simple, no interruption, universal

✗ Requires 2-person technique optimally

2

Supraglottic Airway (SGA)

Class IIa

Indication: Alternative to ETI; non-expert provider

✓ Faster insertion, less CPR interruption

✗ May not protect against aspiration fully

3

Endotracheal Intubation (ETI)

Class IIa

Indication: Experienced provider; prolonged arrest

✓ Definitive airway, continuous CPR possible

✗ Requires skill, significant pause risk

Ventilation During Cardiac Arrest

Without Advanced Airway	With Advanced Airway	Tidal Volume Target	FiO_2 During CPR
30:2 (CPR cycle)	**10 breaths/min**	**~500 mL**	**100%**
2 breaths per 30 compressions	1 breath every 6 seconds	Visible chest rise only	Max O_2 until ROSC

AVOID HYPERVENTILATION — Increases intrathoracic pressure → decreases venous return → reduces coronary perfusion pressure

- Passive oxygenation during initial phase of arrest may be considered (Class IIb)
- After ROSC: target SpO_2 94–98% — avoid hyperoxia (worsens reperfusion injury)
- After ROSC: target $PaCO_2$ 35–45 mmHg — avoid hypocapnia
- Waveform capnography confirms ETT placement (Class I) — mandatory 2026

POCUS in Cardiac Arrest — 2026 Integration

Class I

Cardiac Tamponade

Pericardial effusion → urgent pericardiocentesis

Class I

Tension Pneumothorax

Absent lung sliding → needle decompression

Class I

ETT Confirmation

Lung sliding confirms tracheal (not esophageal) placement

Class IIa

PE Recognition

RV dilation/McConnell sign → thrombolysis consideration

Class IIb

Cardiac Motion

Standstill = poor prognosis; may guide TOR

Class IIb

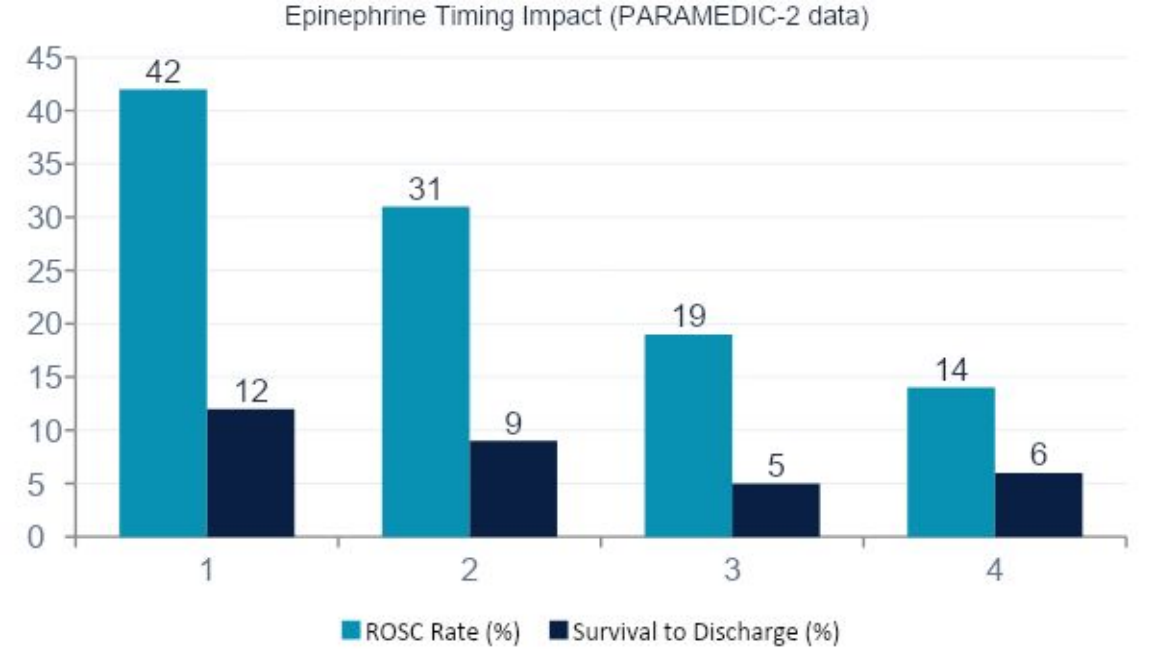
Volume Status / IVC

Collapsible IVC → fluid responsiveness

Pharmacology in Cardiac Arrest

Drug	Indication	Dose	Route	Class 2026
Epinephrine	All cardiac arrest	1 mg q3–5 min	IV/IO	Class I
Amiodarone	Shock-refractory VF/pVT	300 mg → 150 mg	IV/IO	Class IIb
Lidocaine	Shock-refractory VF/pVT	1.5 mg/kg → 0.75 mg/kg	IV/IO	Class IIb (NEW: equivalent)
Adenosine	SVT (ACLS)	6 mg → 12 mg → 12 mg	Rapid IV	Class I
Atropine	Symptomatic bradycardia	0.5 mg q3–5 min (max 3mg)	IV/IO	Class IIa
Dopamine	Bradycardia if atropine fails	2–20 mcg/kg/min	IV infusion	Class IIb
Norepinephrine	Post-arrest hypotension	0.1–0.5 mcg/kg/min	Central IV	Class IIa
Sodium Bicarb	Hyperkalemia/TCA OD	1 mEq/kg	IV	Class IIb (selective)
Calcium Chloride	Hyperkalemia / CCB OD	1g (10 mL of 10%)	IV	Class IIa

Epinephrine — Evidence & Timing



- Early epinephrine (within 5 min) improves ROSC for non-shockable rhythms
- For VF/pVT: withhold epi for first 2 cycles; give after 2nd shock
- Epi improves ROSC but modest effect on neurological survival
- 2026 Update: First dose within 5 min of arrest — Class I, LOE B-NR
- High-dose epinephrine (>1mg) is NOT recommended — harmful outcomes

** Neurological outcomes with epinephrine remain controversial. 2026 guidelines maintain Class I recommendation based on ROSC benefit.*

Amiodarone vs Lidocaine — 2026 Update

Feature	Amiodarone	Lidocaine
Class (2026)	Class IIb	Class IIb (NEW — equivalent)
Indication	Shock-refractory VF/pVT	Shock-refractory VF/pVT
1st Dose	300 mg IV/IO bolus	1.5 mg/kg IV/IO
2nd Dose	150 mg IV/IO	0.75 mg/kg IV/IO (max 3 doses)
Timing	After 3rd defibrillation	After 3rd defibrillation
ROSC Rate	Marginally higher vs placebo	Comparable to amiodarone
Good Neuro Outcome	No significant improvement	No significant improvement
Side Effects	Hypotension, bradycardia	CNS toxicity (dose-dependent)
Availability	Universal	Less consistent globally
Cost	Higher	Lower

2026 Key Message: Either drug is acceptable. Use whichever is available. Neither improves long-term survival — shock quality and CPR matter most.

Post-Cardiac Arrest Care (PCAC)

Hemodynamic

- MAP ≥65 mmHg (target 80–100 mmHg if possible)
- Avoid vasopressor-induced hypertension >120 mmHg
- Dopamine/Norepinephrine for refractory hypotension
- Emergent PCI if STEMI or suspicious OHCA

Respiratory

- SpO_2 94–98% (avoid hyperoxia after ROSC)
- $PaCO_2$ 35–45 mmHg (normocapnia)
- Head-of-bed 30° elevation
- Early extubation when neurologically appropriate

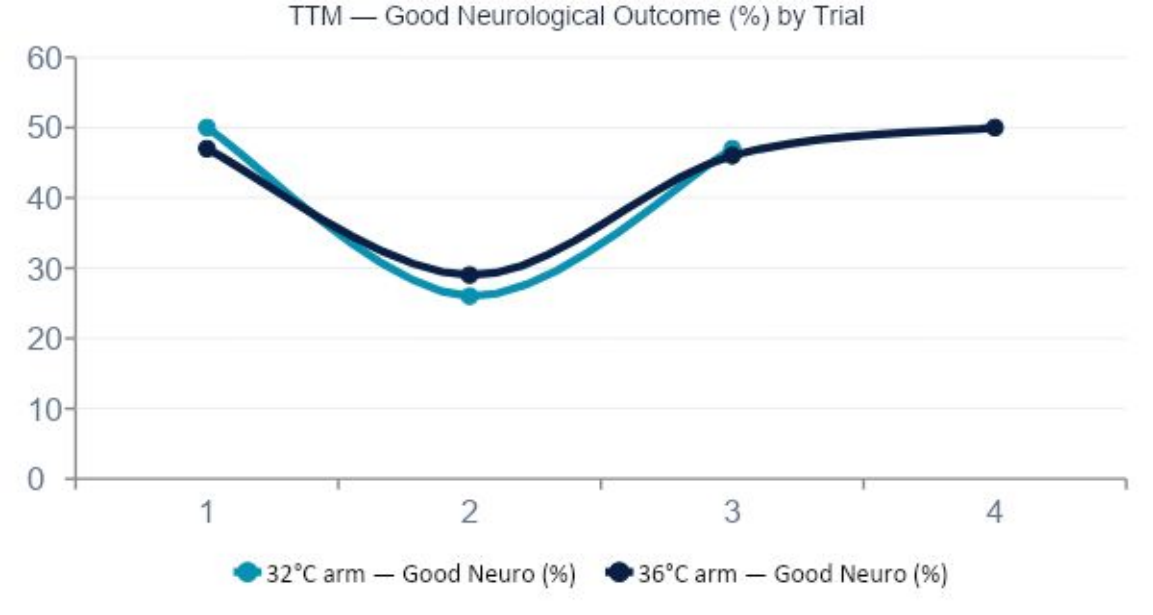
Neurological

- TTM at 36°C for 24h — Class IIa (2026 revised)
- Prevent fever (>37.7°C) for at least 72h
- Continuous EEG monitoring in comatose patients
- Multimodal neuroprognostication ≥72h post-arrest

Metabolic

- Glucose: 140–180 mg/dL (avoid hypoglycemia)
- Correct electrolytes: K^+, Mg^{2+}, Ca^{2+}
- Avoid hypotonic fluids
- Lactate clearance as perfusion marker

Targeted Temperature Management (TTM) — 2026



- 2026 Target: 36°C for minimum 24h (Class IIa)
- No proven benefit of 32°C over 36°C (TTM2 trial)
- Strict fever prevention (>37.7°C) for 72h post-arrest
- Prehospital cooling with cold IV fluids: NOT recommended
- Rewarming: 0.25–0.5°C/hr (avoid rapid rewarming)

Neuroprognostication: Do NOT decide on withdrawal of care before 72h. Use multimodal approach: clinical exam + EEG + SSEP + biomarkers (NSE) + imaging.

Coronary Angiography & PCI Post-Arrest

Situation	Recommendation	Class	Timing
OHCA + STEMI	Immediate angiography & PCI	Class I	ASAP (<90 min)
OHCA + no ST elevation, hemodynamically unstable	Emergent angiography	Class IIa	Within 2h
OHCA + no ST elevation, hemodynamically stable	Delayed angiography (24–72h)	Class IIa (NEW)	24–72h post-arrest
IHCA + suspected ACS	Early angiography	Class IIa	Within 24h
OHCA + obvious non-cardiac cause	No routine angiography	Class III (Harm)	—

COACT Trial & TOMAHAWK Trial: Immediate angiography in stable OHCA without STEMI showed NO benefit vs delayed — current guidance reflects this (Class IIa, 24–72h).

- Patients should receive TTM and hemodynamic support regardless of PCI timing
- Dual antiplatelet therapy: ASA + P2Y12 inhibitor (ticagrelor preferred over clopidogrel)
- Anticoagulation: Unfractionated heparin during PCI; bivalirudin as alternative

Special Situations in Resuscitation

Drowning

- Prioritize ventilation — 5 initial rescue breaths
- C-spine precautions only if mechanism suggests
- Water rescue CPR if safe to do so
- Warm and dry victim — hypothermia likely

Pregnancy

- Left lateral uterine displacement (LUD) — manually
- Standard ACLS drugs are safe in pregnancy
- Perimortem C-section if >20 wks gestation
- Call neonatal team early

Hypothermia

- No asystole confirmation until core temp >30°C
- Withhold epinephrine until temp >30°C
- Active rewarming (warm IV, ECMO if needed)
- Max 3 shocks until rewarmed >30°C

Anaphylaxis Arrest

- Epinephrine 1mg IV (same as standard ACLS)
- Aggressive fluid resuscitation
- Remove trigger if possible
- Methylene blue for refractory vasodilation

Opioid OD

- Naloxone 0.4–2 mg IM/IV/IN — repeat PRN
- Standard CPR if no response to naloxone
- Respiratory support takes priority
- For fentanyl OD: higher naloxone doses may be needed

PE Arrest

- Suspect in PEA + recent long travel/DVT/PICC
- Alteplase 50 mg IV bolus — continue CPR 60–90 min
- ECPR if available
- Surgical thrombectomy / catheter-directed thrombolysis

ECPR (ECMO-CPR) — Expanded 2026 Indications

INCLUSION Criteria

- Age ≤70 years (relative — consider case-by-case)
- Witnessed arrest with bystander CPR
- Initial shockable rhythm (VF/pVT)
- Refractory to ≥3 defibrillations
- CPR initiated within 5 min of collapse
- Estimated reversible cause (ACS, PE, poisoning)
- No asystole >20 min without ROSC

EXCLUSION Criteria

- Prolonged no-flow time >10 min
- Severe comorbidities (end-stage disease)
- Non-survivable trauma/injury
- DNR/advance directive
- Aortic regurgitation/dissection (relative)
- Bilateral femoral access not feasible

ECPR Survival: 20–40% with good neurological outcomes in selected patients. Time from arrest to ECMO cannulation should be <60 min. Class IIa, LOE C-LD.

Bradycardia & Tachycardia Algorithms

Condition	Definition	1st Line Treatment	2nd Line	If Unstable
Symptomatic Bradycardia	HR <60 bpm + symptoms	Atropine 0.5 mg IV	Transcutaneous pacing / Dopamine/Epi infusion	Transvenous pacing
Narrow QRS Tachycardia (SVT)	HR >150 + regular + narrow	Vagal maneuvers → Adenosine 6mg IV	Beta-blocker / CCB	Synchronized cardioversion 50–100J
Wide QRS VT (stable)	HR >150 + wide complex	Amiodarone 150mg IV over 10 min	Procainamide/Sotalol	Synchronized cardioversion 100J
A-Fib (rapid)	Irregular + HR >100	Rate control: Metoprolol/Diltiazem	Rhythm control (if <48h)	Cardioversion if unstable
Polymorphic VT (Torsades)	Irregular wide QRS, twisting	Mg SO_4 2g IV bolus	Isoproterenol, overdrive pacing	Unsynchronized shock if pulseless

Team Dynamics in Resuscitation

Team Leader	Compressor	Airway	IV/IO	Recorder
Directs the resuscitation	High-quality CPR	BVM / advanced airway	Establishes vascular access	Time-stamps all events
Makes treatment decisions	Switch every 2 min	Monitors SpO_2 & $ETCO_2$	Administers medications	Tracks drug doses/timing
Delegates and monitors	Announces fatigue early	Confirms ETT placement	Documents drug timing	Communicates to team
Ensures closed-loop communication	Monitors compression feedback	Ventilation rate 10/min	Draws blood samples	Prepares debrief notes

Closed-Loop Communication & Debriefing

1

SEND message

Team leader gives clear, directed order: 'John — give 1mg epinephrine now'

2

RECEIVE & REPEAT

Receiver reads back: 'Giving 1mg epinephrine — confirmed'

3

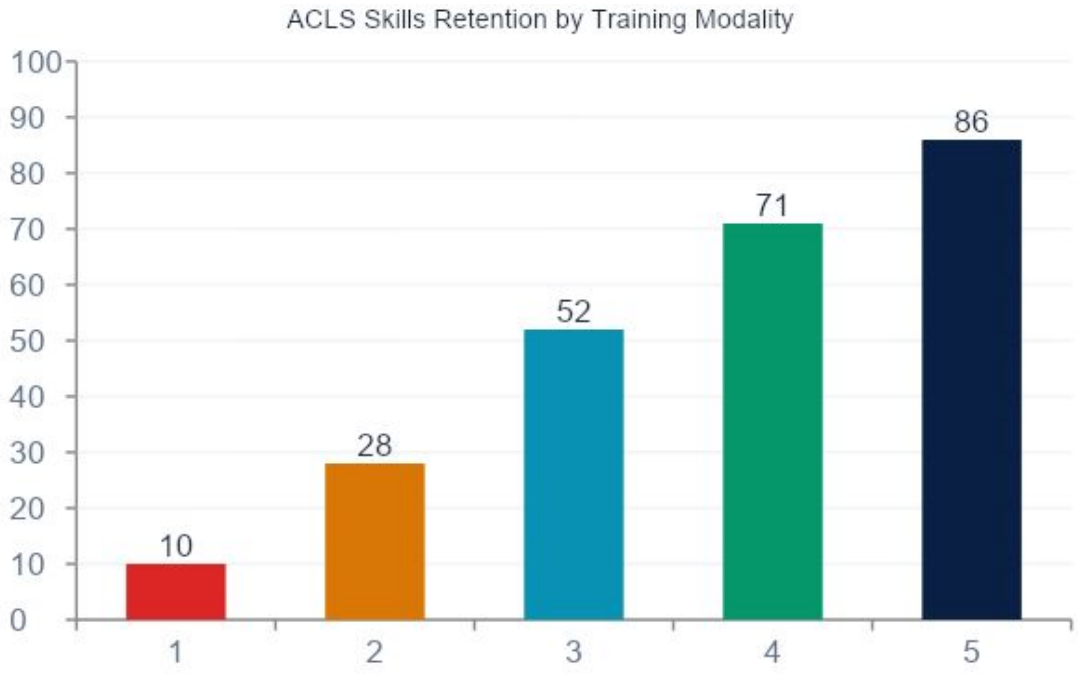
ACKNOWLEDGE

Leader acknowledges: 'Correct — thank you'

Post-Resuscitation Debriefing

- Hot debrief: immediately post-event (5–10 min)
- Cold debrief: structured review 24–48h later
- Focus on system/process — not blame
- Review CPR metrics, drug timing, communication
- Psychological safety for all team members

Simulation-Based ACLS Training



- AHA recommends ACLS recertification every 2 years
- High-fidelity simulation improves team performance by 40%
- Deliberate practice with feedback: most effective method
- Virtual/mixed-reality simulation: emerging modality (2026)
- CPR feedback devices should be used in all training
- Spaced repetition beats massed practice for skill retention

Key Takeaways — ACLS 2026

01

CPR Quality is #1

Rate 100–120/min, depth 2–2.4", full recoil, CCF >80%, ventilation 10/min — these matter more than any drug

02

Early Defibrillation

Every 1 minute delay in defibrillation reduces survival ~10%. AED within 3–5 min is the most impactful intervention

03

Lidocaine = Amiodarone

Both are Class IIb for shock-refractory VF/pVT. Use whichever is available. Neither improves long-term survival

04

TTM at 36°C

Fever prevention is mandatory ≥72h. Lower temperatures (32°C) provide no additional benefit based on current evidence

05

Epinephrine Timing

First dose within 5 min (Class I). For VF/pVT: withhold until after 2nd shock. High doses are harmful

06

POCUS & Capnography

Both are now Class I in 2026. Capnography for ROSC detection; POCUS for reversible causes identification

Thank You

ACLS Update 2026 — Dr. Mehta

Questions & Discussion Welcome

AHA 2026 Guidelines | Dr. Mehta | Evidence-Based

Saving Lives Through Knowledge & Skill